SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I C.P. Elton Pejo (19652603) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	1	1	1	1
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	1	1	1	1

Total Marks 31 /40

Signature of the Faculty

of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Q5. A multimedia application transpers a 600 mb Video file.

The Session Layer Inserts a synchronation checkpoint after every 60mB of transmitted data. During transmission, a failure occurs after 390 mB has been transmitted. Determine the number of checkpoints created before failure and Calculate the amount of data that must be transmitted after recovery. Explain the importance of Synchronization in session layer.

INTRODUCTION

The OSI model consist of seven layers that help computers Communicate over a network. The Session layer is the fifth Layer of the OSI model.

Its main function is to establish, manage, synchronize and terminate communication sessions between Applications.

Checkpoints are transfered or used during long transfers to Record progress.

If a failure occurs, and transfer can resume from the last Checkpoint instead of starting again.

Given data,

Total File Size, 600mB

Checkpoint Interval = 60mB

Data transferred before failure = 390mB

Calculate the number of Checkpoints

$$\text{Checkpoints} = \frac{\text{Data transferred}}{\text{Interval}}.$$

$$= \frac{390}{60} = 6.5.$$

Only Complete checkpoints are counter.

The checkpoints are 60, 120, 180, 240, 300, 36.

∴ Checkpoints = 6.

• Calculate retransmitted data

The last Completed checkpoint is at 360mB

Retransmitted data = 390 - 360

= 30MB.

30MB must be retransmitted.

IMPORTANCE OF SYNCHRNOTION.

- Recovery : Helps Recover communication after a failure.
- Time Saving : Avoids Restarting the entire transfer
- Bandwidth Saving : Reduces unnecessary dat transmission.
- Efficiency : Helps large file transfer more Efficientiy
- Session Management : Helps Coordinate Communication

between applications.

EXAMPLE :

Suppose a student uploads a large Video file. If the connection fails after 390mB, the last checkpoint is at 360 MB.

The transfer can Recover from that Checkpoint. and the 30mB sent after it may need to be transmitted.

CONCLUSION

The Session Layer Manages Communication Sessions and Supports Synchronization. In this problem, 6 checkpoints are Completed and 30mB must be retransmitted after the failure. Checkpoints save time and reduce unneccary data transmission.

Q.6 Before transmission the presentation layer Compresses a File by 40%. The Compressed data is encrypted, add 5% overhead to the compressed size. Calculate the Compressed File Size, the encrypted File Size, and tot percentage Reduction Compared to the Original Size Explain how the presentation Layer improves transmis efficiency and Security.

INTRODUCTION

- The Presentation Layer is the Sixth layer of the OSI model
- It helps Represent data in a format that the receiving application can understand.
- Its main funtions include data transmission, translation, compression and encryption.
- Compression Reduces file size.
- Encryption protects information from unauthorised user.

Given Data

Orginal file size = 900mB

Compression = 40%.

Encryption Overhead = 5%.

Calculate the Compressed Size

After 40% of compression, 60% of Orginal size Remains

Formula

$$\text{Compressed Size} = 900 \times \frac{60}{100}$$

$$= 540\text{MB}.$$

Therefore the Compressed size is 540mb.

Calculate Encryption.

$$\text{Overhead} = 540 \times \frac{5}{100}$$

$$= 27\ \text{MB}$$

Calculate the Final Size

$$\text{Final Size} = 540 + 27$$

$$= 567\ \text{MB}.$$

Calculate the percentage Reduction

Calculate the Size Saved

$$900 - 567 = 333\ \text{MB}.$$

$$\text{Percentage Reduction} = \frac{333}{900} \times 100$$

$$= 37\ \%$$

Advantages of Compression and encryption.

- Compression:

  Reduces file size and save storage.

- Faster Transfer:

  Smaller files can take less time to transmit

- Bandwidth Saving:

  Reduces the amount of data sent through a network.

- Security:

  Encryption helps protect Confidential information.

- Data transmission.

CONCLUSION

The Presentation Layer Supports data representation and is commonly associated with Compression and encryption. In this problem the orginal 900 mB file becomes 540 mB after Compression.

- Encryption adds 27mB producing a final size of 567 mB.

- The Overall Size reduction is 27%.

A Packet experiences the following processing delays at different OSI Layers ; Application layer = 4ms. transport layer = 3ms , Network layer = 5ms , Data link layer = 2ms , physical layer = 1ms. The propogation delay IS 18 ms and the transmission delay 12 ms. Calculate the total end to end delay experienced by the packet and how each layer contributes to a communication process.

INTRODUCTION

- The osi model Consist of seven layers that explain how data is transfered between Components.
- Each layer performs a specific function during Communication.
- End-to-End delay is a total time taken for data to travel from the sender to the oleciver. It includes processing delay, transmission delay and propogation delay.
- Processing delay is the time taken to examine and process data.
- Transmission delay is the time taken or required to place the packets bits onto the Network link.

Given data

Application Layer = 4ms

Transport Layer = 3ms

Network Layer = 5ms.

Data Link layer = 2ms

Physical layer = 1ms

Propagation delay = 18ms

Transmission delay = 12ms.

Calculate total Processing delay

Processing delay = 4+3+5+2+1

= 15ms.

Therefore total processing delay = 15mls.

Calculate total End-End delay

Total delay = Processing delay + transmission delay + Propogation delay

15 + 12 + 18

= 45ms.

The total end-end delay is 45 milli seconds.

functions of the OSI Layers Involved

- Application Layer :

  proride Network services.

- Transport Layer :

  Supports Communication.

- Network layer :

  Handles logical addressing and routing packets

- Data Link layer.

  Transfers data in frames.

- Physical Layer

  Transmits raw bits through Cables.

CONCLUSION

End-to-End delay measures the time aequired For data to travel from a sender to a aeciver. In this problem the total Processing delay is 15ms, the transmission delay is 12ms and propogation delay is 18ms.

3. A 18,000 byte IP packet is transmitted through a router where the maximum Transmission Unit is 1500 bytes. Assuming Each Fragment requires a 20 byte IP header Calculate the total number of fragments created and the total header overhead Introduced. Explain how ~~flow control~~ the network layers ensures successful delivery of fragmented packets to the destination.

INTRODUCTION

• The Network Layer is the third layer of the OSI model. It is responsible for logical addressing and routing packets From the Source to the destination.

• Different network links support different maximum packet sizes. The maximum Transmission Unit specifies the maximum size of the packet.

• When an IPv4 packet is larger than the MTU. It may be divided into smaller pieces called fragments If Fragmentation is permitted. Each fragment contains its own IP header. The destination uses the fragment information to reassemble the orginal Packet.

Given data

Original IP packet size = 12,000 bytes.

MTU = 1,500 bytes.

IP header Size = 20 bytes.

Calculate the pay load per fragment.

The MTU includes both IP header and payload.

payload = MTU - Header = 1500 - 20

= 1480 bytes.

Original payload

Original payload = 12000 - 20

= 11980 bytes

Calculate number of fragments.

$$\frac{11980}{1480} \approx 8.09$$

8 Fragments

Calculate total header Overhead

Total Overhead = 8×20

= 160 bytes.

Steps Involved in Fragmentation

- The Network checks the packet Size against MTU
- The Original payload is dividend into smaller portions
- Each fragment recives its own IP header.
- Fragmentation information identifies the Orginal packet

Advantages of Fragmentation.

- Allows eligible IPv4 Packets to pass through links with Smaller MTU's.
- Support Communication across networks with different packet Size.
- Helps divide large packets into smaller pieces when Required.